

Alexander Skinderev

925-819-7790 a.skinderev@gmail.com linkedin.com/in/askinderev github.com/SashaSkind

Education

University of California, Santa Cruz GPA: 3.65/4

Expected Graduation: **June 2027**

Bachelor of Science in Computer Science

- **Relevant Coursework:** DSA (C++), Probability & Statistics in CS, Machine Learning, Linear Algebra w/Computational Applications, Computer Architecture

Experience

Bright Pattern – AI-powered contact center platform serving 500+ enterprise customers globally.

Dec 2024 – Present

Junior Software Engineer

San Bruno, CA

- Reduced speech-to-text API calls by **50%** and raised transcription **accuracy 15%** by shipping a production dialogue-level pipeline that merges multi-party audio into one contextual stream with Silero VAD before inference; extended **Arabic** completeness **30%** by giving the model full conversational context, recovering short context-dependent utterances the per-speaker path dropped.
- Freed over **40 hours per month** of QA capacity, cut regression cycle time **95%**, and grew suite coverage from **0 to 32%** in 5 weeks by architecting an agentic test-automation system using agent orchestration and **Model Context Protocol (MCP)** servers to drive the product end-to-end and originate real SIP phone calls.
- Surfaced **6 previously-undocumented product behaviors** and increased silent-failure detection **4×** by designing a 5-layer verification framework that grounds every run against UI state, call routing, transcription metadata, transcript content, and backend persistence, catching cases a single-signal assertion scored as passing.

Projects

Idiolect – 2nd place, JacHacks SF (Gemma 4 Challenge); on-device speech recognition that learns a speaker's own vocabulary

- Cut error on personal terms **87%** (R-WER 0.593 → 0.074) and overall WER **69%** after 17 user corrections, using contextual biasing over a correction-harvested vocabulary instead of per-speaker fine-tuning, so personalization needs zero training and no labelled audio.
- Recovered n-best hypotheses from **Parakeet TDT 0.6B** (MLX) by rebinding the beam-search collapse step from installed source at import time, keeping top-1 bit-identical while preserving alternatives; a **4B open-weight model** reranks candidates against user vocabulary and escalates to its native audio path only when they disagree (R-WER 0.185 → 0.074). Runs entirely on-device with no hosted API.
- Built the evaluation suite (WER/R-WER, calibration, input-difficulty sweep, TORGO simulation) and **published a negative result:** no measurable gain on real dysarthric speech, with analysis showing that corpus's phonetically-balanced prompts leave the mechanism almost nothing to bias toward.

Language Classifier – multilingual language identification, deployed on Hugging Face Spaces

- Fine-tuned **XLM-RoBERTa** on 90K samples across **20 languages** to **99.6%** held-out accuracy, well past classical baselines (Naive Bayes 86.1%, logistic regression 89.2%), and served it behind FastAPI at **~17 ms** warm latency on Apple Silicon.

DailyGate – Hackathon winner; self-improving agent with earned-autonomy governance; published on Guild's agent marketplace

- Built an agent that **earns autonomy** through a **Bayesian Beta-Binomial trust model** (30-day time-decay, confidence gating, dynamic per-risk thresholds): approvals and overrides update a per-category posterior that promotes or demotes the agent across permission tiers, where each tier is a separate sandboxed agent with distinct tool grants behind a classifying router rather than a prompt-level rule.
- Shipped end-to-end with a multi-tenant trust service (each org earns autonomy in isolation from zero), instant demotion on a single override, hard-capped high-stakes categories, and live decision tracing through Langfuse.

ColdReach – Zero to Agents (Vercel) hackathon winner; AI cold-outreach product, public beta at trycoldreach.app

- Built with one teammate and continued developing into public beta. Shipped a production agent that runs a multi-turn tool-use loop over a contact-enrichment API and drafts three tone variants in parallel on a second model, cutting user time-to-send **85%** and per-chat inference cost **70%** through dual-model routing.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C/C++, Go, SQL, Swift, R | **Spoken:** English, Russian

ML & Inference: PyTorch, Hugging Face Transformers, MLX, Ollama, open-weight and on-device model serving, fine-tuning (XLM-RoBERTa, DistilRoBERTa), multilingual NLP, speech-to-text (Parakeet TDT, Whisper, Silero VAD), n-best decoding and beam search, contextual biasing, reranking, model cascading, real-time inference

Evaluation & Statistics: WER/R-WER benchmarking, ROC/AUC, calibration, difficulty sweeps and ablations, external-dataset reporting, statsmodels (OLS regression, experimental design with controls), Bayesian inference

Agents & Infra: Model Context Protocol (MCP), Vercel AI Gateway, multi-turn tool-use, agent orchestration, model routing, Langfuse, FastAPI, Hono, Next.js, React, ClickHouse, Neon Postgres, Docker, Fly.io, AWS, Git, Linux